

Endpoint Parity Loss in a Bessel Wronskian: an exact obstruction to kernel-and-anchor proofs of global ratio monotonicity

Lluis Eriksson

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Abstract

For $\beta > 0$ write $I_m = I_m(\beta)$ and define

$$a_m = I_m^2((m-1)I_{m-1}^2 + (m+1)I_{m+1}^2), \quad b_m = mI_m^4,$$

with sine series $F_A(t) = \sum_{m \geq 1} a_m \sin(mt)$ and $F_B(t) = \sum_{m \geq 1} b_m \sin(mt)$. The associated Wronskian $\mathcal{W} = 2(F'_A F_B - F_A F'_B)$ is negative exactly when F_A/F_B is decreasing. We isolate what can and cannot be proved from two natural inputs: the Neumann convolution kernel $I_0(2\beta \sin(\phi/2))$ and the small-coupling anchor $(-4096/\beta^{10})\mathcal{W} \rightarrow 4\sin^3 t$.

First, the convolution does prove the denominator statement $F_B(t) > 0$ for $0 < t < \pi$. Second, the endpoint is governed by two alternating quantities, c_3 and B_π , through the exact law $\mathcal{W}(\pi - d) = -4c_3 B_\pi d^3 + O(d^5)$. We prove $B_\pi > 0$, derive integral representations, and establish the logarithmic cancellation law

$$\lim_{\beta \rightarrow \infty} -\frac{1}{\beta} \log \frac{|c_3 B_\pi|}{c_3^{\text{abs}} B_\pi^{\text{abs}}} = 8 - 4\sqrt{2}.$$

Thus a positive-term majorant loses an exponentially decisive factor. Finally, we give a smooth one-parameter perturbation theorem: one may keep F_B and its kernel unchanged, preserve positivity and strict coefficient-ratio ordering, and preserve every jet at $\beta = 0$, while choosing either sign of the endpoint cubic coefficient. Consequently those structural data, even taken together, do not imply global Wronskian negativity. This is a no-go theorem for a proof architecture, not a counterexample to the Bessel conjecture. A short high-precision kill-test accompanies the paper; no computer-assisted inequality is used in the proofs.

1 The question and the status of the result

The following ratio-monotonicity statement arises after an exact Wronskian reduction of a surface-positivity problem.

Conjecture 1 (Bessel Wronskian conjecture). For every $\beta > 0$,

$$F_B(t) > 0, \quad \left(\frac{F_A}{F_B} \right)'(t) < 0 \quad (0 < t < \pi).$$

Equivalently, $F_B > 0$ and $\mathcal{W} < 0$ on $(0, \pi)$.

The first assertion will be proved exactly below. The second assertion is *not* proved here. Our main result instead separates a valid part of the proposed kernel route from a fatal loss at $t = \pi$. The distinction is important: the perturbations in Theorem 5 do not obey all modified-Bessel recurrences, and hence do not refute Conjecture 1.

The Wronskian identity itself is elementary. If $F_x(t) = \sum_{m \geq 1} x_m \sin(mt)$, then

$$2(F'_a F_b - F_a F'_b) = \sum_{m < n} (a_m b_n - a_n b_m) [(m - n) \sin((m + n)t) + (m + n) \sin((n - m)t)]. \quad (1)$$

It follows by expanding the products and antisymmetrizing. Throughout, all Bessel series and their differentiated series converge locally uniformly; the rearrangements below are therefore legitimate.

2 The convolution kernel closes $F_B > 0$

The Neumann addition theorem gives the two parity forms

$$K_+(\phi) = I_0\left(2\beta \cos \frac{\phi}{2}\right) = \sum_{m \in \mathbb{Z}} I_m(\beta)^2 e^{im\phi}, \quad (2)$$

$$K_-(\phi) = I_0\left(2\beta \sin \frac{\phi}{2}\right) = \sum_{m \in \mathbb{Z}} (-1)^m I_m(\beta)^2 e^{im\phi}. \quad (3)$$

For periodic convolution normalized by $(f * g)(t) = (2\pi)^{-1} \int_0^{2\pi} f(s)g(t - s) ds$, both squares give

$$P_4 := K_+ * K_+ = K_- * K_- = \sum_{m \in \mathbb{Z}} I_m(\beta)^4 e^{imt}, \quad F_B = -\frac{1}{2} P'_4. \quad (4)$$

Lemma 2 (strict circular convolution). *If q is a nonconstant, even, continuous function on the circle that is strictly decreasing with geodesic distance from 0, then $q * q$ is strictly decreasing on $(0, \pi)$ and $(q * q)' < 0$ there whenever q is smooth.*

Proof. Use the layer-cake representation of $q - \min q$ by centered circular arcs. The convolution is a positive superposition of overlap lengths of two such arcs whose centers are separated by t . Each overlap length is nonincreasing for $0 < t < \pi$. At any fixed t there is an open set of pairs of level arcs whose endpoints meet in the unsaturated regime; on this set the derivative of the overlap is strictly negative. Strict decrease of q makes this set have positive level measure. Differentiation is allowed for smooth q and yields the pointwise derivative claim. $\square$

Theorem 3. *For every $\beta > 0$, $F_B(t) > 0$ for $0 < t < \pi$.*

Proof. On $[0, \pi]$, $K_+(t) = I_0(2\beta \cos(t/2))$ is strictly decreasing, because $I'_0(x) = I_1(x) > 0$ for $x > 0$. Apply Lemma 2 to (4): $P'_4(t) < 0$, and hence $F_B(t) = -P'_4(t)/2 > 0$. $\square$

This proof uses the kernel at full strength and contains no alternating majorization. The obstruction begins with the numerator and becomes sharp at the opposite endpoint.

3 The exact small-coupling anchor

The fixed-order Taylor expansions of I_m give

$$\begin{aligned} a_1 &= \frac{\beta^6}{128} + O(\beta^8), & b_1 &= \frac{\beta^4}{16} + O(\beta^6), \\ a_2 &= \frac{\beta^6}{256} + O(\beta^8), & b_2 &= \frac{\beta^8}{2048} + O(\beta^{10}). \end{aligned}$$

All remaining modes enter later. Substitution in the Wronskian yields, uniformly for $t \in [0, \pi]$,

$$\mathcal{W}(t; \beta) = -\frac{\beta^{10}}{1024} \sin^3 t + O(\beta^{12}), \quad -\frac{4096}{\beta^{10}} \mathcal{W}(t; \beta) \longrightarrow 4 \sin^3 t. \quad (5)$$

Thus the parabolic anchor has the desired sign for every fixed interior t when β is small. Uniformity of the absolute error in (5), however, does not supply a relative estimate near $t = \pi$, where the leading term has a triple zero.

4 The endpoint coefficient

Put $d = \pi - t$. The linear coefficient of $F_A(\pi - d)$ vanishes by an edgewise telescoping:

$$\sum_{m \geq 1} (-1)^{m+1} m a_m = 0. \quad (6)$$

Indeed, the two appearances of $I_m^2 I_{m+1}^2$ have coefficients

$$(-1)^{m+1} m(m+1) \quad \text{and} \quad (-1)^{m+2} (m+1)m,$$

which cancel. Define

$$c_3(\beta) = \frac{1}{6} \sum_{m \geq 1} (-1)^{m+1} m(m+1)(2m+1) I_m^2 I_{m+1}^2, \quad (7)$$

$$B_\pi(\beta) = \sum_{m \geq 1} (-1)^{m+1} m^2 I_m^4. \quad (8)$$

Taylor expansion, followed by the same edge collection used in (6), gives

$$F_A(\pi - d) = c_3 d^3 + O(d^5), \quad F_B(\pi - d) = B_\pi d + O(d^3), \quad (9)$$

and consequently

$$\mathcal{W}(\pi - d) = -4c_3 B_\pi d^3 + O(d^5), \quad \lim_{t \uparrow \pi} \frac{-\mathcal{W}(t)}{\sin^3 t} = 4c_3 B_\pi. \quad (10)$$

The denominator coefficient has a positive integral. From $P_4''(\pi) = 2B_\pi$, differentiating the convolution and integrating by parts once gives

$$B_\pi = \frac{\beta^2}{\pi} \int_0^{\pi/2} \sin u \cos u I_1(2\beta \cos u) I_1(2\beta \sin u) du > 0. \quad (11)$$

Therefore the desired endpoint sign is exactly the nontrivial statement $c_3(\beta) > 0$.

For later use, set $f(x) = I_1(x)/2$, $y = 2\beta \sin u$ and $z = 2\beta \cos u$. Applying (3), differentiating the correlation, and integrating by parts gives the half-range identity

$$c_3 = \frac{1}{6\pi} \int_0^{\pi/2} J(u, \beta) du, \quad (12)$$

where

$$J = -z^2 f'(y) f'(z) + zy^2 f'(y) f''(z) + yf(y) f'(z). \quad (13)$$

The integrand is not pointwise positive. This is not a numerical accident: for every sufficiently large β , its negative boundary region and positive saddle region are both nonempty. Hence (12) is a signed cancellation identity rather than a positive-kernel proof of $c_3 > 0$.

5 The exponential parity loss

Introduce the positive envelopes

$$c_3^{\text{abs}} = \frac{1}{6} \sum_{m \geq 1} m(m+1)(2m+1) I_m^2 I_{m+1}^2, \quad (14)$$

$$B_\pi^{\text{abs}} = \sum_{m \geq 1} m^2 I_m^4. \quad (15)$$

Theorem 4 (endpoint cancellation exponent). *As $\beta \rightarrow \infty$,*

$$c_3(\beta) = \frac{\beta^{3/2} e^{2\sqrt{2}\beta}}{24 \cdot 2^{1/4} \pi^{3/2}} (1 + O(\beta^{-1})), \quad (16)$$

$$B_\pi(\beta) = \frac{\beta^{1/2} e^{2\sqrt{2}\beta}}{4 \cdot 2^{3/4} \pi^{3/2}} (1 + O(\beta^{-1})). \quad (17)$$

Moreover

$$\lim_{\beta \rightarrow \infty} \frac{1}{\beta} \log c_3^{\text{abs}} = \lim_{\beta \rightarrow \infty} \frac{1}{\beta} \log B_\pi^{\text{abs}} = 4, \quad (18)$$

and therefore

$$\boxed{\lim_{\beta \rightarrow \infty} -\frac{1}{\beta} \log \frac{|c_3 B_\pi|}{c_3^{\text{abs}} B_\pi^{\text{abs}}} = 8 - 4\sqrt{2}.} \quad (19)$$

Proof. We give the details needed to identify both the sign and the exponential scale. In (12), the phase is $y + z = 2\beta(\sin u + \cos u)$. It has a unique nondegenerate maximum at $u_0 = \pi/4$. Write $x = \sqrt{2}\beta$ and $u = u_0 + v$. Uniformly for $|v| \leq \beta^{-2/5}$, the fixed-order large-argument expansion and its differentiated forms give

$$f^{(j)}(y) = \frac{e^y}{2\sqrt{2\pi y}} (1 + O(\beta^{-1} + |v|)), \quad j = 0, 1, 2,$$

and the analogous statement at z . The leading term in J is $zy^2 f(y) f(z)$; at the saddle it equals

$$\frac{x^2 e^{2x}}{8\pi} (1 + O(\beta^{-1} + |v|)).$$

Also $y + z = 2x - xv^2 + O(\beta v^4)$. Laplace's method in (12) now gives

$$c_3 = \frac{1}{6\pi} \frac{x^2 e^{2x}}{8\pi} \sqrt{\frac{\pi}{x}} (1 + O(\beta^{-1})),$$

which is (16). Outside the displayed central window, $\sin u + \cos u$ has a uniform quadratic deficit until a fixed neighborhood is exited and a uniform positive deficit thereafter; the standard exponential Bessel bound makes that contribution smaller than the stated error. This also justifies the differentiated asymptotics.

The same saddle applied to (11) is simpler. At u_0 the prefactor $\sin u \cos u$ equals $1/2$, and $I_1(x)^2 = e^{2x} (2\pi x)^{-1} (1 + O(x^{-1}))$. The Gaussian factor $\sqrt{\pi/x}$ yields (17).

For (18), any fixed positive summand gives the lower exponential rate 4, since $I_m(\beta) = e^\beta (2\pi\beta)^{-1/2} (1 + O(\beta^{-1}))$ for fixed m . For the upper bound, the generating function

$$\sum_{m \in \mathbb{Z}} I_m(\beta) z^m = \exp\left(\frac{\beta}{2}(z + z^{-1})\right)$$

and the three-term recurrence control every fixed absolute polynomial moment. Positivity and repeated product bounds dominate (14) and (15) by a polynomial in β times $e^{4\beta}$. Taking logarithms proves (18). Combining the two signed rates $2\sqrt{2}$ with the two absolute rates 4 proves (19). $\square$

The rate $8 - 4\sqrt{2} = 2.3431457505\dots$ is the kill-test for any positive-term or absolute-value barrier. A barrier whose slack decays only algebraically cannot resolve the endpoint sign at large β .

6 A structural no-go theorem

We now show that the loss is logical, not merely a weakness of a particular estimate. Let $p = (2, 1, 0, \dots)$. It satisfies

$$\sum_{m \geq 1} (-1)^{m+1} m p_m = 0, \quad -\frac{1}{6} \sum_{m \geq 1} (-1)^{m+1} m^3 p_m = 1. \quad (20)$$

Theorem 5 (structural non-identifiability). *Let $(A_m(\beta), B_m(\beta)) = (a_m(\beta), b_m(\beta))$ be the Bessel arrays above, and suppose the known strict ordering $A_m/B_m < A_{m+1}/B_{m+1}$ is included among the structural inputs. There are two families $\tilde{A}^+$ and $\tilde{A}^-$ such that:*

1. *B is unchanged; hence the convolution representation and $F_B > 0$ are unchanged;*
2. *every $\tilde{A}_m^\pm$ is positive and $\tilde{A}_m^\pm/B_m$ remains strictly increasing in m for every $\beta > 0$;*
3. *$\tilde{A}^\pm - A$ is flat at $\beta = 0$, so the two families have the same complete small- β asymptotic expansion and in particular the same anchor (5);*
4. *at one common $\beta = \beta_*$, the endpoint cubic coefficients have opposite signs. Thus their Wronskians have opposite signs in a punctured left neighborhood of $t = \pi$.*

Consequently items 1–3 cannot imply global Wronskian negativity.

Proof. For a scalar function h , put

$$\tilde{A}_m = A_m + h(\beta)p_m, \quad \tilde{B}_m = B_m.$$

By (20), the endpoint linear cancellation is preserved and the cubic coefficient becomes $\tilde{c}_3 = c_3 + h$.

Choose a large center $\beta_* = R$. Theorem 4 shows $|c_3(R)| = O(R^{3/2}e^{2\sqrt{2}R})$. In contrast, the fixed-mode coefficients A_1, A_2, B_1, B_2 have exponential scale e^{4R} times an inverse power of R . Direct insertion of the fixed-order Bessel expansions gives, for fixed m ,

$$\frac{A_{m+1}}{B_{m+1}} - \frac{A_m}{B_m} = \frac{4(2m+1)}{R^2} + O(R^{-3}).$$

Multiplying these gaps by the relevant B_m shows that the admissible size of a perturbation in modes 1, 2 has exponential scale e^{4R} times an inverse power of R . Therefore, for all sufficiently large R , there is an $\eta_R > |c_3(R)|$ such that $|h(R)| < \eta_R$ preserves positivity and both affected ratio inequalities.

Fix $0 < \varepsilon < \eta_R - |c_3(R)|$ and, for $\sigma > 0$, set

$$\psi_{R,\sigma}(\beta) = \exp\left(-\frac{1}{\beta^2} - \frac{(\beta - R)^2}{\sigma^2} + \frac{1}{R^2}\right), \quad h_\pm(\beta) = (-c_3(R) \pm \varepsilon)\psi_{R,\sigma}(\beta).$$

Here $\psi(R) = 1$; it is real analytic on $(0, \infty)$ and extends by 0 to a C^∞ function flat at 0. At R we have $\tilde{c}_3^\pm(R) = \pm\varepsilon$.

It remains only to choose σ . In a neighborhood of R , strict positivity and the two strict ratio gaps persist by continuity and the choice of η_R . On compact sets separated from that neighborhood, $\psi_{R,\sigma} \rightarrow 0$ uniformly as $\sigma \downarrow 0$. Near 0 it is smaller than every power of β , whereas the coefficients and their strict gaps have finite algebraic orders. Near infinity its Gaussian decay dominates the fixed-mode coefficient scales and gap denominators. Thus a sufficiently small σ preserves positivity and ratio order for all $\beta > 0$. Flatness gives item 3, and (10) gives item 4. $\square$

Remark 6 (exact scope). The theorem rules out any argument whose hypotheses see only the B -kernel, positivity of the coefficients, strict ordering of A_m/B_m , and the full small-coupling jet. A successful proof of Conjecture 1 must use additional Bessel structure that rejects the perturbation, for example an exact signed identity, recurrence-compatible differential inequality, or a cancellation-preserving certificate.

7 Reproducible kill-test

The companion script `scripts/wronskian_endpoint_kill_test.py` is a diagnostic, not an interval proof. It uses 180-decimal arithmetic, truncates well beyond the active Bessel modes, and checks the exact normalizations before measuring the cancellation. A representative run is:

β	endpoint ratio	$\log(\text{ratio})$	next fitted rate
8	$3.86868117 \cdot 10^{-5}$	-10.1600118	2.1068310
16	$1.85217583 \cdot 10^{-12}$	-27.0146600	2.2301610
32	$5.90114568 \cdot 10^{-28}$	-62.6972361	2.2878524
64	$9.45545116 \cdot 10^{-60}$	-135.9085142	—

Here “endpoint ratio” is $|c_3 B_\pi|/(c_3^{\text{abs}} B_\pi^{\text{abs}})$. The fitted rates approach $8 - 4\sqrt{2}$. At $\beta = 32$ and $d = 10^{-4}$, the quotient of $\mathcal{W}(\pi - d)/d^3$ by $-4c_3 B_\pi$ is 1.0000009109773. The finite perturbation that flips c_3 has relative size $c_3/A_1 = 1.270694641 \cdot 10^{-12}$ and preserves positivity and the checked ratio order.

The acceptance checks use explicit exceptions rather than Python `assert`; the same run is required under ordinary Python and `python -O`. The option `-self-test-mutations` injects a false predicate and requires the harness to reject it. None of these floating-point observations is used to establish Theorems 3, 4, or 5.

8 Implications for proof design

The two proposed legs now have sharply different outcomes.

1. The convolution kernel is strong enough to prove $F_B > 0$ globally. It should be retained as the denominator lemma.
2. The parabola $4\sin^3 t$ is an exact small- β anchor but is not a global barrier by itself. Any continuation in (t, β) must carry an endpoint variable normalized by $c_3 B_\pi$, not merely absolute Fourier moments.

3. The large- β endpoint window requires signed resolution on the scale $e^{-(8-4\sqrt{2})\beta}$ relative to the naive positive envelope. Polynomially accurate structural bounds necessarily miss it.
4. Coefficient-ratio order is useful but logically insufficient for sine-series ratio monotonicity. Theorem 5 remains true even after adjoining all jets at $\beta = 0$.

A plausible surviving architecture is therefore modular: use the exact convolution for F_B ; isolate $c_3 > 0$ by a cancellation-preserving signed identity; prove a normalized endpoint window after factoring $\sin^3 t$; and bridge that window to a compact interior region. The present paper does not claim that this program has been completed.

9 Priority and limitations

The addition formulas (2)–(3) are classical Neumann–Graf identities; standard sources are Watson’s treatise and the NIST Digital Library of Mathematical Functions [3, 6]. The large-argument expansions with differentiable error control are also classical [4, 5]. Ratio criteria for ordinary power series go back at least to Biernacki and Krzyż [2]; they do not directly apply to sine series. Positivity and monotonicity phenomena for Bessel sums have a substantial earlier literature, including Askey–Steinig [1].

The contribution claimed here is narrower: the exact endpoint reduction, the cancellation exponent (19), and the perturbative non-identifiability theorem for the specified proof architecture. We do not claim novelty for the classical identities or Laplace method. We also do not claim a proof or disproof of Conjecture 1. Related programme manuscripts record the Wronskian reduction and a separate certificate-based route to the global inequality [7, 8]; the present result is useful precisely because it explains why a much shorter kernel-and-anchor proof cannot close without a new signed lemma.

Data and verification statement

The source, diagnostic script, invocation commands, output transcript, byte counts, and SHA-256 hashes are distributed with the manuscript. The paper proofs are ordinary analytic arguments. No Lean build, interval oracle, or sustained remote computation was used for this paper. The numerical kill-test is labelled diagnostic and is reproducible on a single CPU process.

References

- [1] R. Askey and J. Steinig, Some positive trigonometric sums, *Transactions of the American Mathematical Society* **187** (1974), 295–307. doi:[10.1090/S0002-9947-1974-0338481-3](https://doi.org/10.1090/S0002-9947-1974-0338481-3).
- [2] M. Biernacki and J. Krzyż, On the monotonicity of certain functionals in the theory of analytic functions, *Annales Universitatis Mariae Curie-Skłodowska, Sectio A* **9** (1955), 5–23. [UMCS digital repository record](#).
- [3] NIST Digital Library of Mathematical Functions, §10.44(ii), addition theorems for modified Bessel functions, dlmf.nist.gov/10.44.ii, accessed 4 August 2026.
- [4] NIST Digital Library of Mathematical Functions, §10.40, asymptotic expansions for large argument, dlmf.nist.gov/10.40, accessed 4 August 2026.
- [5] F. W. J. Olver, *Asymptotics and Special Functions*, A K Peters, Wellesley, MA, 1997.

- [6] G. N. Watson, *A Treatise on the Theory of Bessel Functions*, 2nd ed., Cambridge University Press, 1944.
- [7] L. Eriksson, A Wronskian identity for antisymmetrized sine sums, programme manuscript, July 2026.
- [8] L. Eriksson, Global ratio monotonicity for a killed von Mises bridge, programme manuscript, July 2026; [ai.vixra:2607.0089](#).